

Tong (Stone) Shi

GIS&Transportation Engineer | [linkedin.com/in/stoneshiucb](https://www.linkedin.com/in/stoneshiucb) | <https://ditt.ai/> | <https://github.com/camel-ai/oasis>

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EDUCATION

University of California, Berkeley

Aug 2023 – May 2024

M.S. in Civil and Environmental Engineering-Transportation Engineering

Departmental Fellowship (\$25000); Student Opportunity Fund (\$1350)

University of California, Santa Barbara

Aug 2019 – Jun 2022

B.A. in Geographical Information Science | Minor in Architecture and Urban History

SKILL

GIS (7yrs) - ArcGIS suite, QGIS, ENVI, API- Google Map, OpenStreetMap, GDAL, AutoCAD, Civil3D

Cloud (2yrs) – AWS, Azure, Google Cloud

Coding (3yrs) - Python (Scikit machine learning, OSMnx, Geopandas), R, Java (MATSim)

RESEARCH EXPERIENCE

Institute of Transportation Studies, UC Berkeley | Project Leader

Jan 2024 – Ongoing

- Led a spatiotemporal distribution analysis of emissions from 66 airports in California to assess environmental and public health impacts on surrounding communities.
- Utilized GIS for mapping and scikit model to reveal the emission dispersion and spatial inequality to inform sustainable airport operation and planning., addressing a gap in research on airport and aircraft emission effects.
- Applied EPA and CalEnviroScreen 4.0 values to evaluate socio-economic vulnerability and identify exposure populations.
- Developed strategies for sustainable airport operations, by integrating safety zone analysis and considering risk evaluations, aiming to inform mitigation and future management practices that enhance the sustainability of airports.

CalTrans & NEXTOR III | Research Assistant

Jan 2024 – July 2024

- Conducted the safety zone accident analysis at 259 airports in California, revealed increased risk compared to 2011 Handbook.
- Utilized datasets including the National Transportation Safety Board (NTSB) accident records to digitize runway and safety zone geometries, employing methods such as point pattern analysis to identify accident clustering near runway lines.
- Employed GIS and geospatial machine learning techniques to calculate the relative distances of accident locations to the nearest safety zone boundaries, visualizing the results through scatter plot maps.
- Provided statistical analyses to pinpoint critical areas for safety improvements to inform future decision-making adjustments.

CAMEL-AI | Transportation System Researcher

Nov 2024 - Ongoing

- Built an LLM-agent system to model passenger responses to transit disruptions and assess multimodal network resilience.
- Implemented traffic coordination AI agents that dynamically adjust transit assignments to optimize urban mobility.
- Integrated PyMATSim, NetworkX, and OSMnx for traffic forecasting, validating behavior against census congestion reports.
- Benchmarked LLM-powered decision-making against traditional simulations to evaluate transit efficiency improvements.

HuMNet Lab, UC Berkeley | Project Leader

Sep 2023 – Apr 2024

- Led a team to enhance cycling experiences by complex network perception analysis at 90,000+ roads in Bogota, Colombia;
- Improved the perception assessment model based on Google Street View, advancing the accuracy and reliability of the system;
- Pioneered approaches in urban infrastructure analysis, contributing to the safer and enjoyable cycling environment in Bogota;
- Research results "'How does my cognition change on the roadway?' Using street view imageries to assess urban perception in human daily mobility" are presented at 2024 American Association of Geographers Annual Meeting.

MesoAmerican Research Center, UCSB | Laboratory Manager

Oct 2021 – April 2023

- Supervised a 9-person GIS team, digitized an hydrological network for the ancient Tikal Site in Guatemala.
- Created the watershed model and verified the locations of 300+ wells using archaeological records and DEM.
- Simulated well-based water supply to estimate supported population and milpa crop yield through rotating cultivation.
- Research results "Visualize Sustainable Land Use in the Tropical Maya Forest: Revitalizing the Old Tikal Map for New Geospatial Analyses" are presented at 2023 American Association of Geographers Annual Meeting.

WORK EXPERIENCE

Ditto | Product Manager, Prompt Engineer

February 2025– Ongoing

- Developed a GIS-driven dating location search system, using Python (GeoPandas), ArcGIS, QGIS, and routing APIs to optimize date venue suggestions based on users' mobility patterns, public transport access, and mutual accessibility zones.
- Designed a preference-exclusion algorithm that removed incompatible matches from the pool, reducing irrelevant suggestions by over 40%, using multi-agent representations of user values, deal-breakers, and compatibility thresholds.
- Edited prompts to train AI agents to match location-based user patterns and scenario planning, supporting real-time accessibility optimization features for the dating experience provided to over 20k users at California.

TossTo.ai | Founder, Product Manager

Nov 2023 – Jan 2025

- Led a team in designing the product prototypes, rolling out a beta version for early users and investors to gather feedback.
- Conceived the innovative concept of the product from over 30 primary users, integrating LLM-driven technology to enhance travel and dining experiences by AI-Agent.
- Designed POI recommendation algorithms for real-time, AI-enhanced user experiences.